Computer Architecture and Operating Systems
COSC 22073/BECS 22233

Lesson 07: Introduction to Operating Systems

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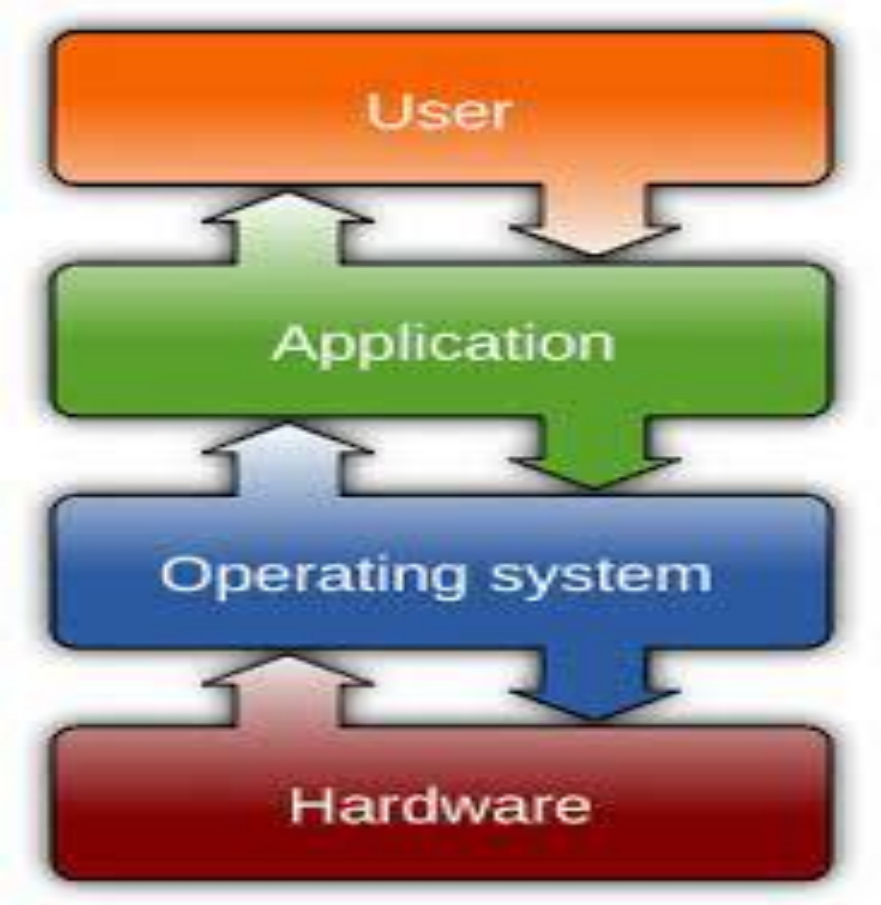
Faculty of Science
University of Kelaniya

Coverage of the Lesson

- Introduction to Different OS
- Process and Threads
- Scheduling
- Memory Management
- File System

Why do we have to study OS?



What is an Operating System

An operating system (OS) is system software that manages computer hardware, software resources, and provides common services for computer programs.

Operating System Goals

- Execute user programs and make solving user problems easier.
- Make the computer system convenient to use
- Use the computer hardware in an efficient manner

Computer System Structure

Computer system can be divided into four components:

1.Hardware – provides basic computing resources

CPU, memory, I/O devices

2.Operating system

Controls and coordinates use of hardware among various applications and users

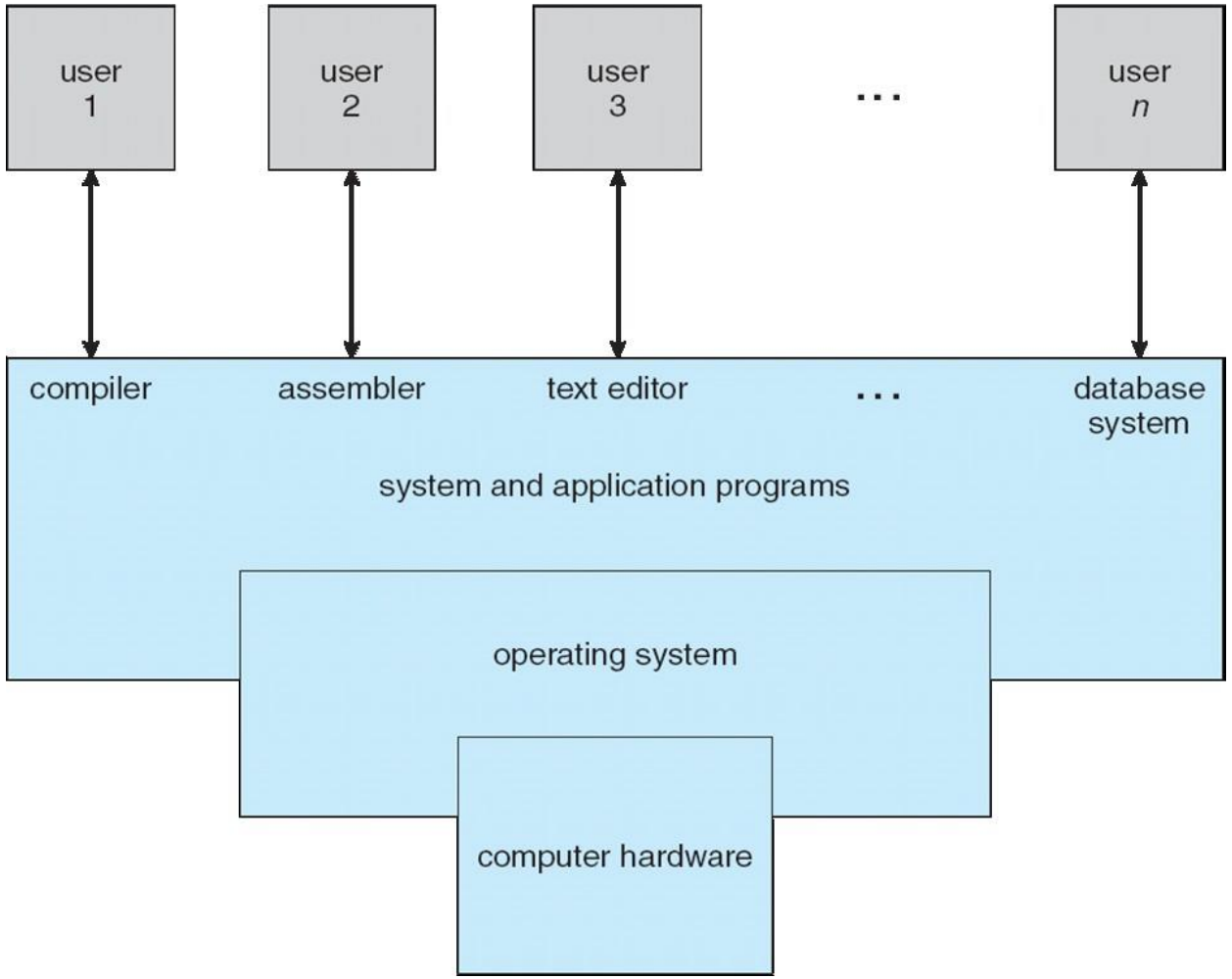
3.Application programs – define the ways in which the system resources are used to solve the computing problems of the users

Word processors, compilers, web browsers, database systems, video games

4.Users

People, machines, other computers

Four Components of a Computer System



Types of Operating Systems

- **Desktops**

Windows, Ubuntu, Mac OS

- **Servers**

Redhat, Ubuntu, Windows Server

- **Mobile OS**

Android, iOS, Ubuntu Touch, Windows Touch

- **Embedded OS**

Contiki OS

- **RTOS**

QNX, VxWorks, RTLinux

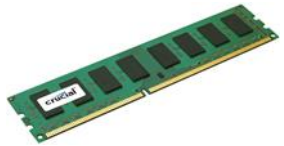
- **Secure OS**

SeLinux, SeL4

Typical OS Architecture

- **Hardware**

Hardware is the physical elements of a computer system. It is the physical parts of a computer, such as the monitor, mouse, keyboard, computer data storage, hard disk drive (HDD), graphic cards, sound cards, memory, motherboard, and so on.



Typical OS Architecture

- **Kernel**

The Kernel is a computer program that manages input/output requests from software, and translates them into data processing instructions for the central processing unit and other electronic components of a computer.

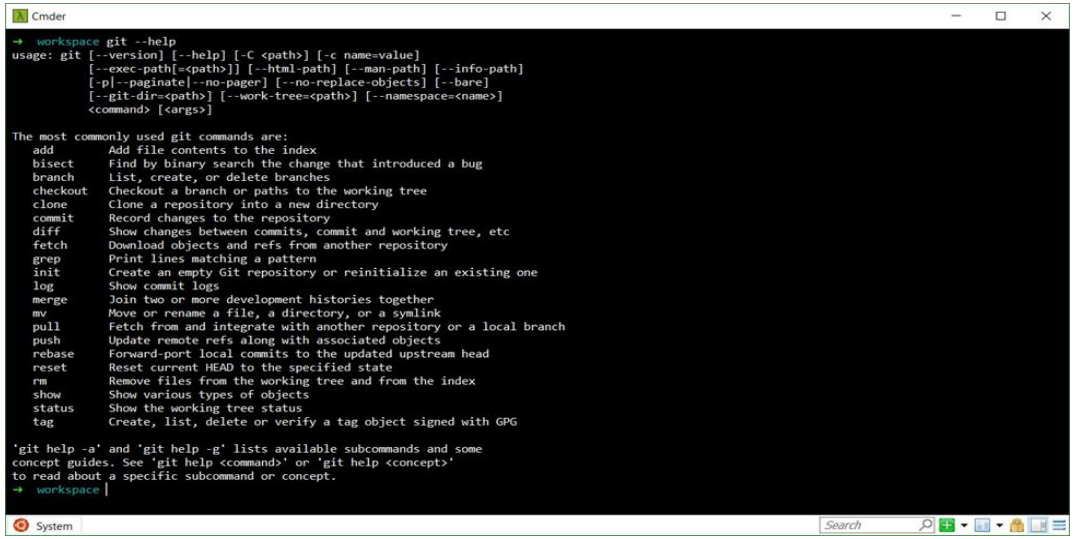
Typical OS Architecture

- **Shell**

The Shell is an interface for access to an operating system's services. In general, operating system shells use either a command-line interface (CLI) or graphical user interface (GUI).

Two Types of Interfaces in OS

CLI (Command Line Interface)



GUI (Graphical User Interface)

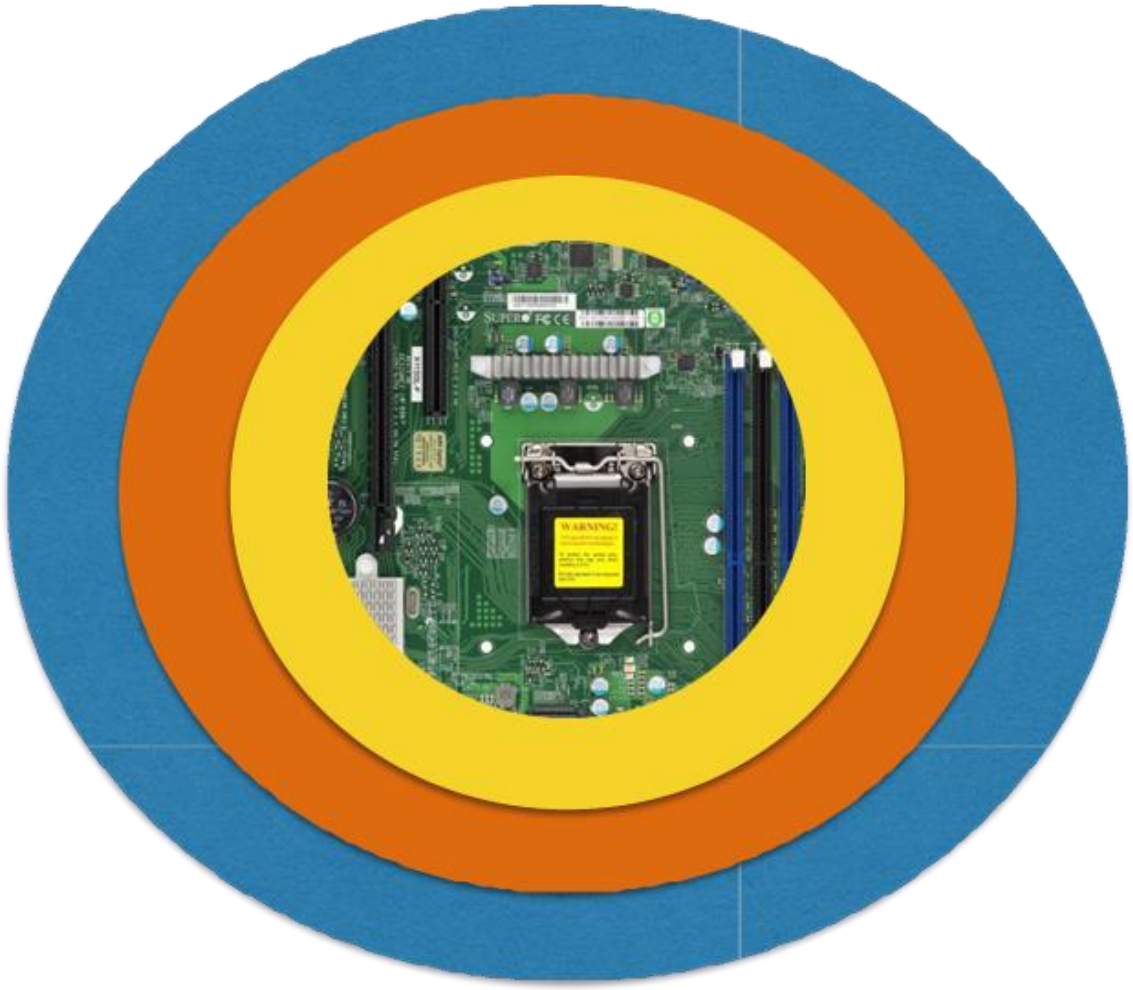
Typical OS Architecture

- **User Applications**

User Applications are computers program designed to perform a group of coordinated functions, tasks, or activities for the benefit of the user.

Typical OS Architecture

- User Applications
- Shell
- Kernal
- Hardware



Introduction to Common Operations Systems

UNIX

- Developed by Ken Thompson and Dennis Ritchie
- Unix was launched in 1969
- It is a CLI (Command-Line Interface)
- Written in the C programming language
- Led to a variety of academic and commercial variants, e.g.

University of California, Berkeley (BSD), Microsoft (Xenix), IBM (AIX) and Sun Microsystems (Solaris)

Introduction to Common Operations Systems

Windows

- Developed by Microsoft - Bill Gates & Paul Allen
- Windows was launched in 1985
- Built on DOS (Disk Operating System), which is a CLI (Command-Line Interface)
- Has two modes User Mode (the user cannot directly access the hardware) and Kernel Mode (the user can access the hardware)
- Runs on Computers, Tablets (Surface), and integration with Xbox

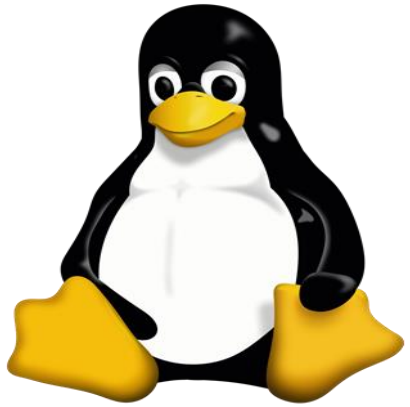
Introduction to Common Operations Systems

Apple Mac OS (Leter OS X)

- Developed by Apple - Steve Wozniak & Steve Jobs
- MacOS was launched in 1984
- MacOS is a native GUI (Graphical User Interface)
- MacOS evolved into OS X, which combined technologies from MacOS, Unix, and NeXT
- Runs on Computers and other devices.

Introduction to Common Operations Systems

Linux



- Developed by Linus Benedict Torvalds
- Linux was launched in 1992
- Linux is a CLI (Command-Line Interface)
- Torvalds made the code of Linux freely available to everyone on the internet, and therefore lots of people created their own versions of Linux, e.g. Debian, RedHat, SUSE, SlackWare, Gentoo, Ubuntu

Introduction to Common Operations Systems

Android

- Developed by Andy Rubin, Rich Miner, Nick Sears, and Chris White
- Android was launched in 2003
- Based on the Linux kernel
- Android is a GUI designed primarily for touchscreen mobile devices such as smartphones and tablets

Introduction to Common Operations Systems

iOS

- Developed by Steve Jobs and Scott Forstall
- iOS was launched in 2007
- Based on the MacOS
- iOS is a GUI designed primarily for touchscreen mobile devices such as iPhones, iPods, iPads, and AppleTV

Two Main Usage of the OS

- Hardware Abstraction

  Turns hardware into something that applications can use.

- Resource Management

  Manage system’s resources

Consider the following program

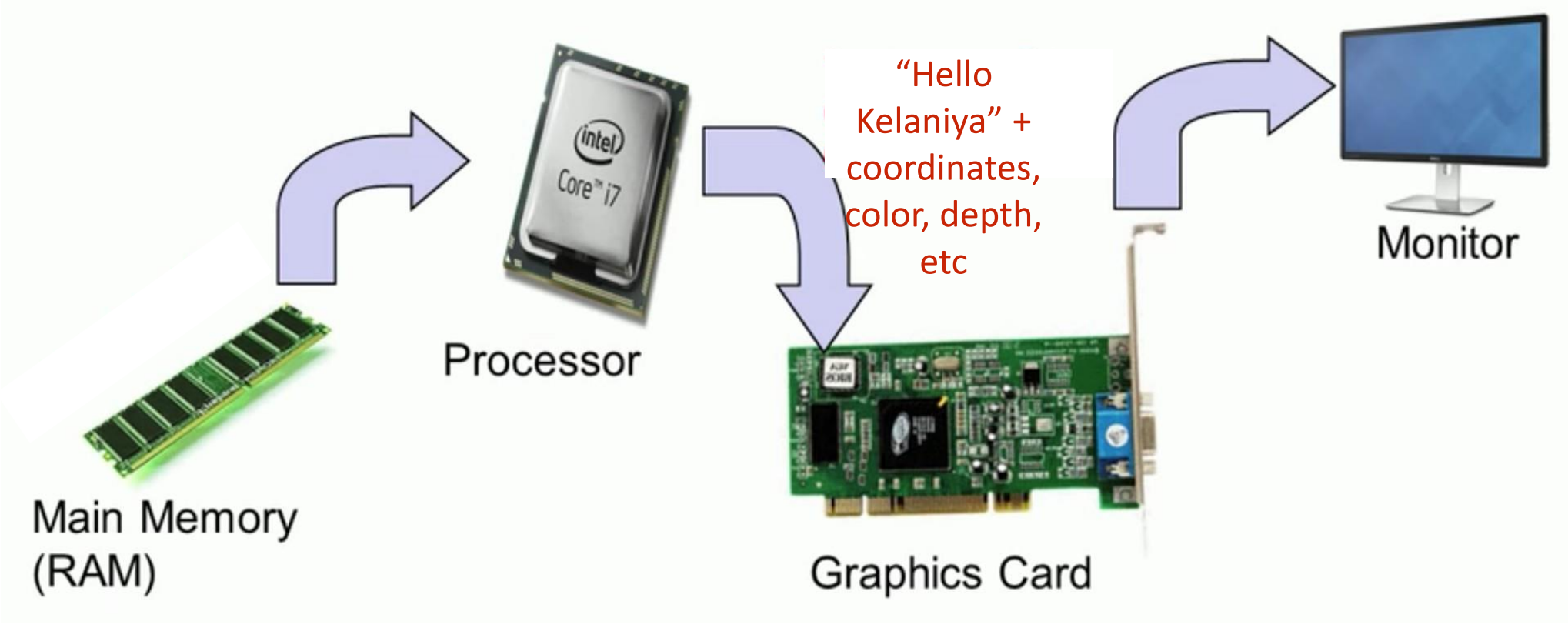
- What is the output of the following program?
- How it is displaying on the screen?

```
#include <stdio.h>

void main()
{
        char str[] = "Hello Kelaniya";
        printf("%s\n",str);
}
```

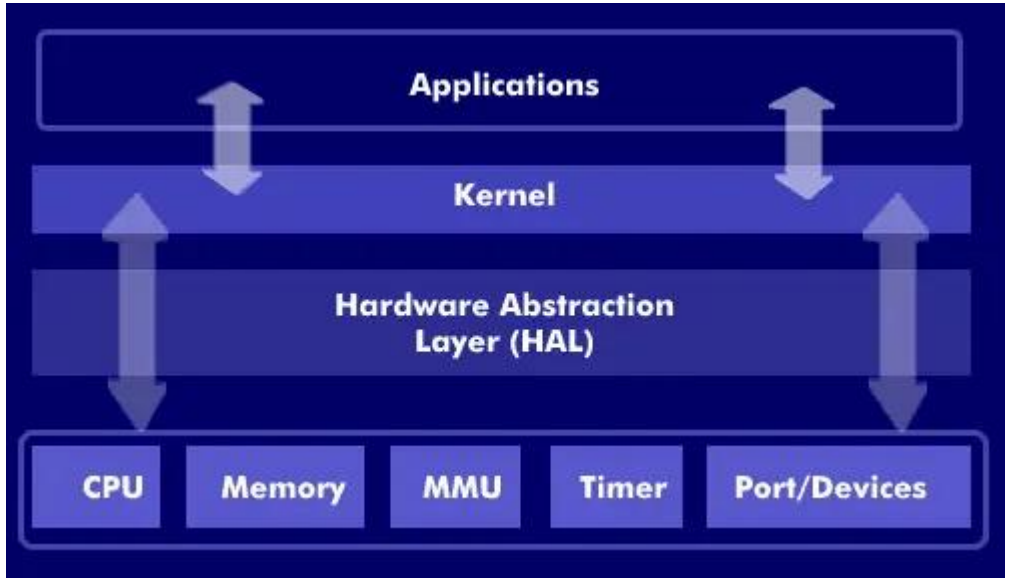
Consider the following program

Displaying the output of the program



Hardware Abstraction

In computers, a hardware abstraction layer (HAL) is a layer of programming that allows a computer OS to interact with a hardware device at a general or abstract level rather than at a detailed hardware level.



Advantages of Hardware Abstraction

- Easy to develop apps
- Reusable functionality
- Portability

OS as a Resource Manager

- Multiple applications but limited hardware resources

```
#include <stdio.h>

void main()
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```
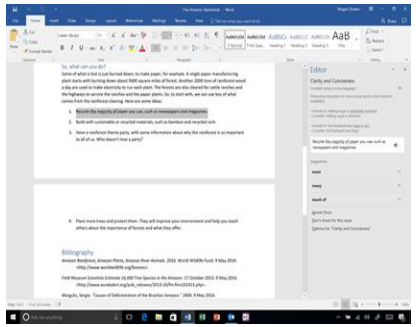



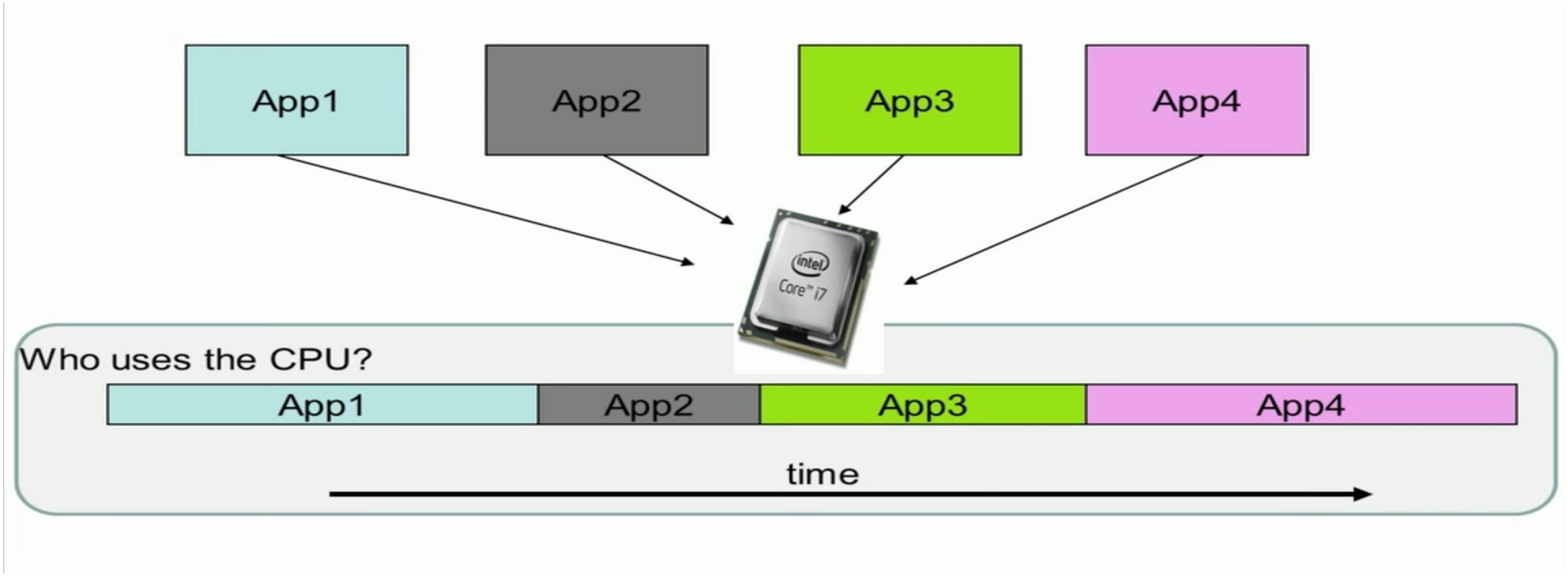
OS as a Resource Manager

OS must Manage:

- CPU
- Memory
- Network
- GPU
- Secondary Storage
- Other hardware resources in the system

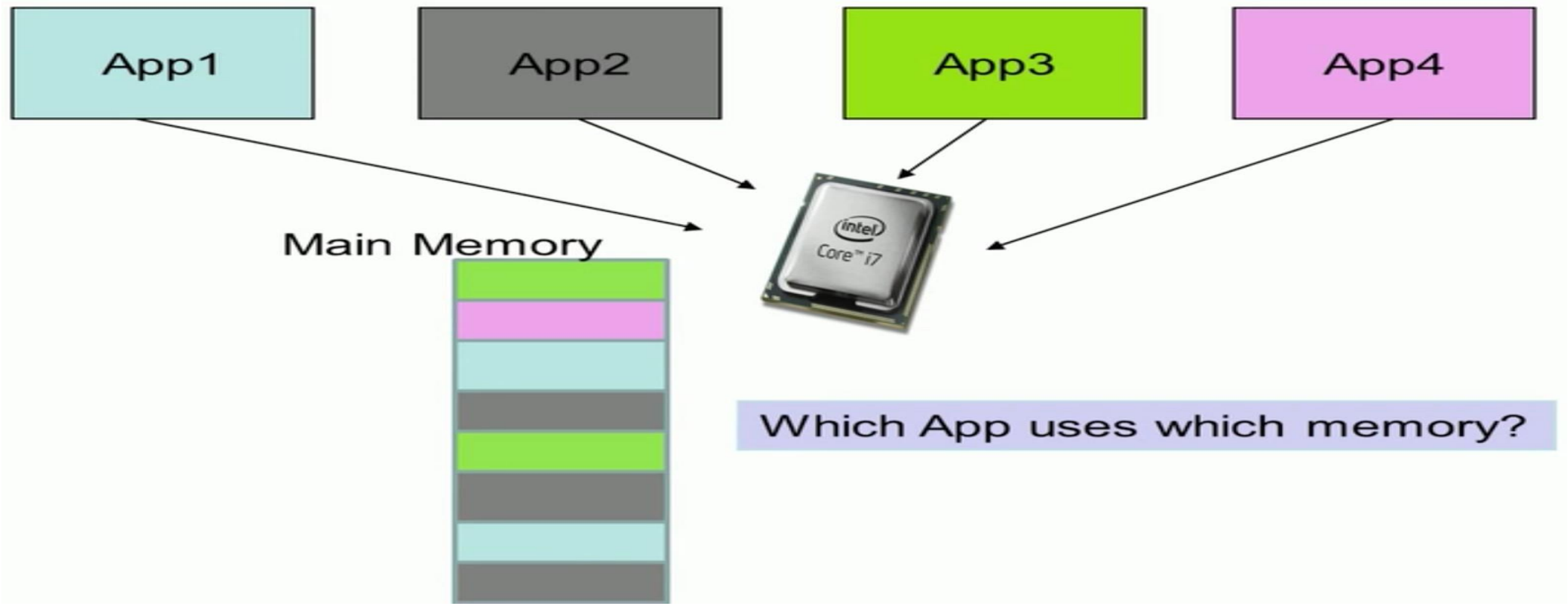
OS as a Resource Manager

OS allows sharing CPU



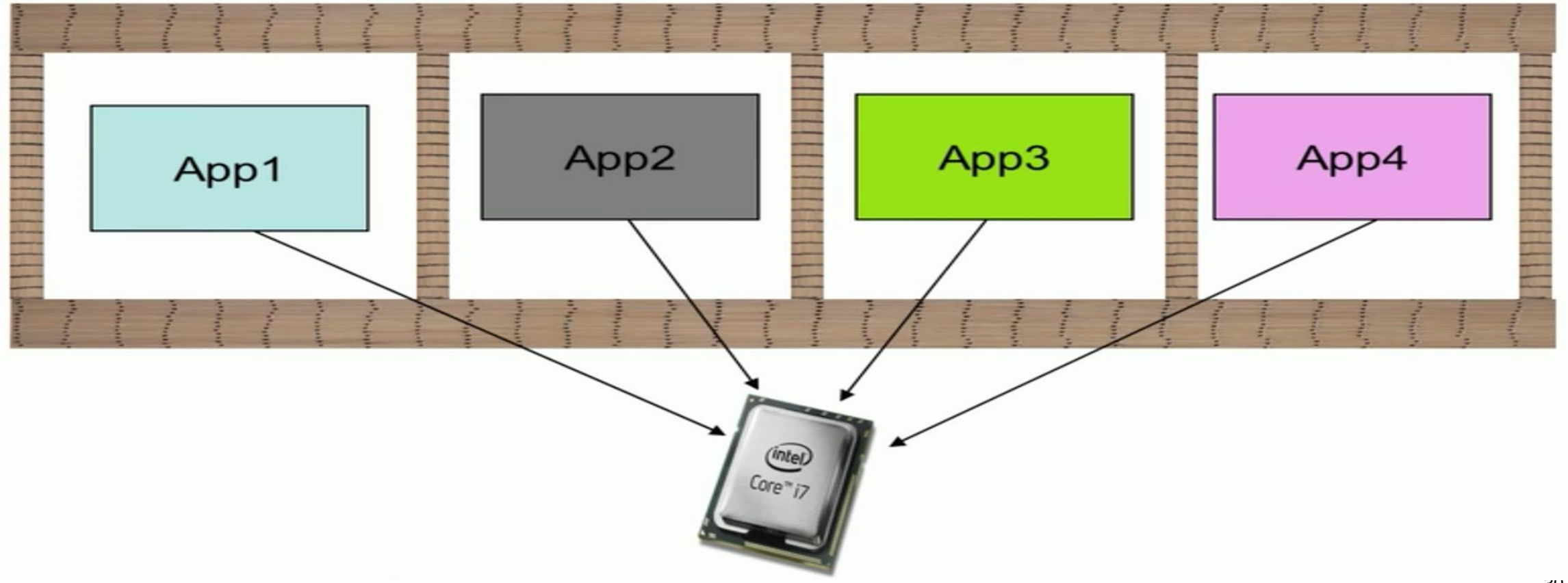
OS as a Resource Manager

OS allows sharing Memory



OS as a Resource Manager

OS allows sharing but isolation



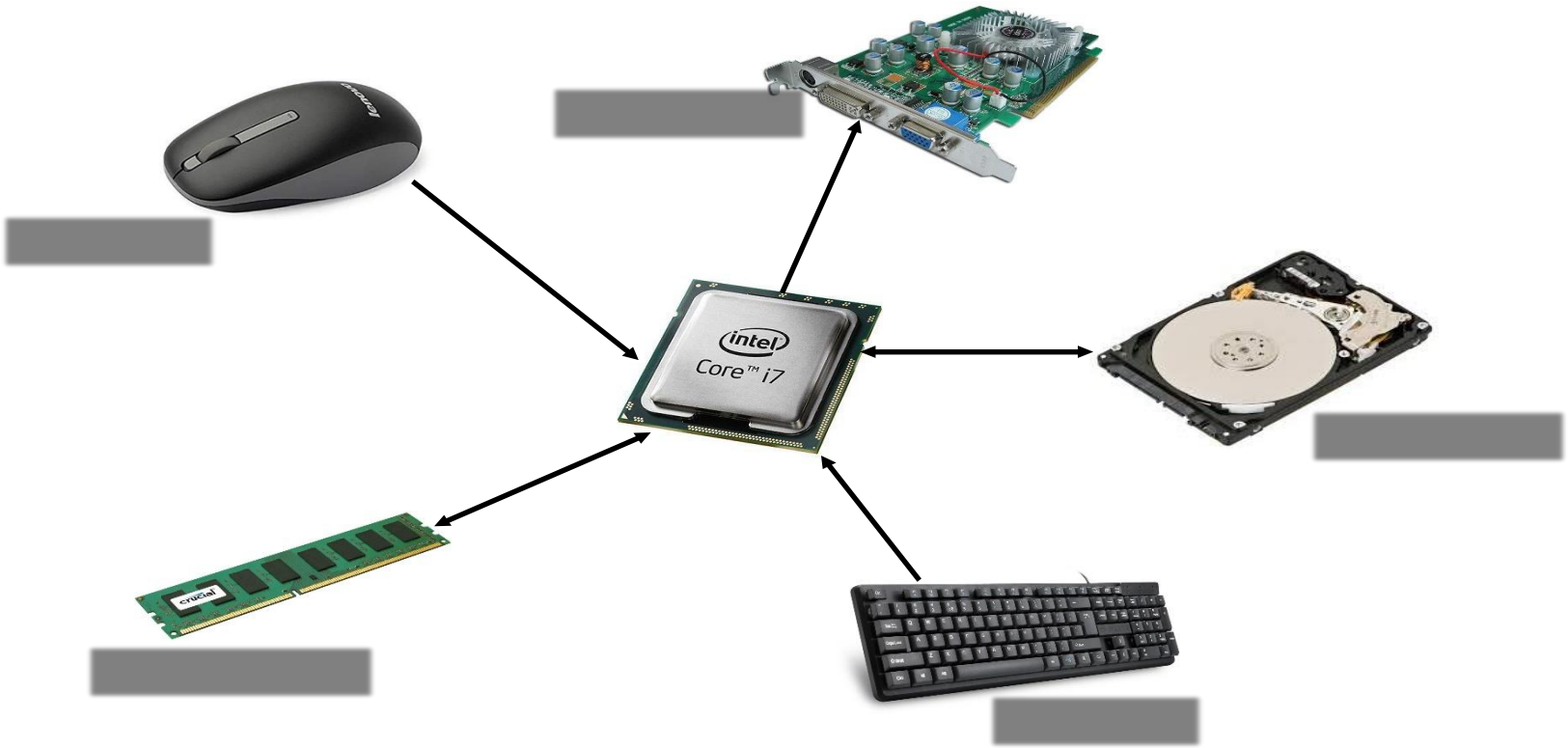
OS as a Resource Manager

Advantages of OS resource management

- Allows multiple applications to share resources
- Protects applications from each other
- Improves performances by efficient utilization of resources

PC Hardware

Everything has an address.



Address Types

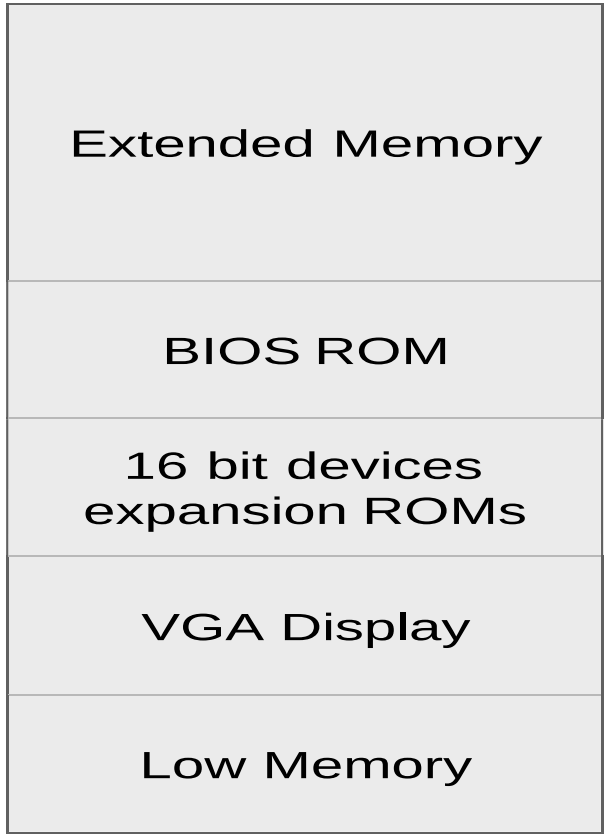
- Memory Addresses
- IO Adresses
- Memory Mapped IO Addresses

Memory Addresses

- Range - 0 to size of the RAM
- Where the main memory is mapped data for code, heap, stack, OS and etc
- Access by load/store instructions

Memory Addresses

How the RAM is organised?



IO Addresses

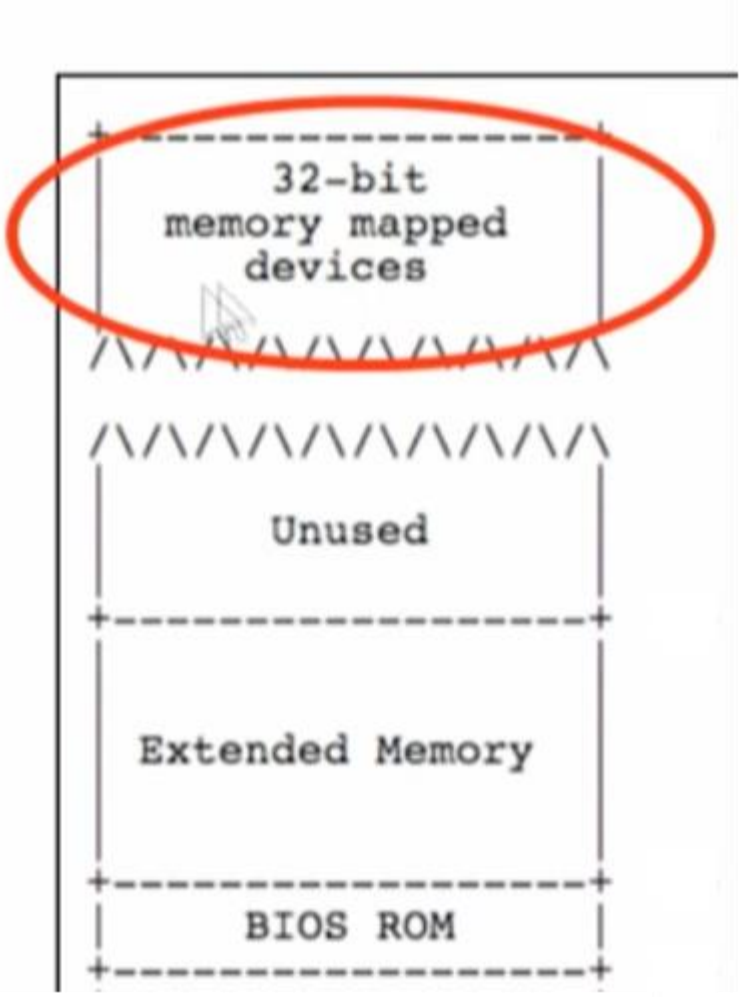
- Range - 0 to 2^16 - 1 (65535)
- Used to access devices
- Access by in/out instructions

I/O address range	Device
00 – 1F	First DMA controller 8237 A-5
20 – 3F	First Programmable Interrupt Controller, 8259A, Master
40 – 5F	Programmable Interval Timer (System Timer), 8254
60 – 6F	Keyboard, 8042
70 – 7F	Real Time Clock, NMI mask
80 – 9F	DMA Page Register, 74LS612
87	DMA Channel 0
83	DMA Channel 1
81	DMA Channel 2
82	DMA Channel 3
8B	DMA Channel 5
89	DMA Channel 6
8A	DMA Channel 7
8F	Refresh
A0 – BF	Second Programmable Interrupt Controller, 8259A, Slave
C0 – DF	Second DMA controller 8237 A-5
F0	Clear 80287 Busy
F1	Reset 80287
F8 – FF	Math coprocessor, 80287
F0 – F5	PCjr Disk Controller
F8 – FF	Reserved for future microprocessor extensions
100 – 10F	POS Programmable Option Select (PS/2)
110 – 1EF	System I/O channel
140 – 15F	Secondary SCSI host adapter
170 – 177	Secondary Parallel ATA Disk Controller
1F0 – 1F7	Primary Parallel ATA Hard Disk Controller

Memory Mapped IO Addresses

- Why?

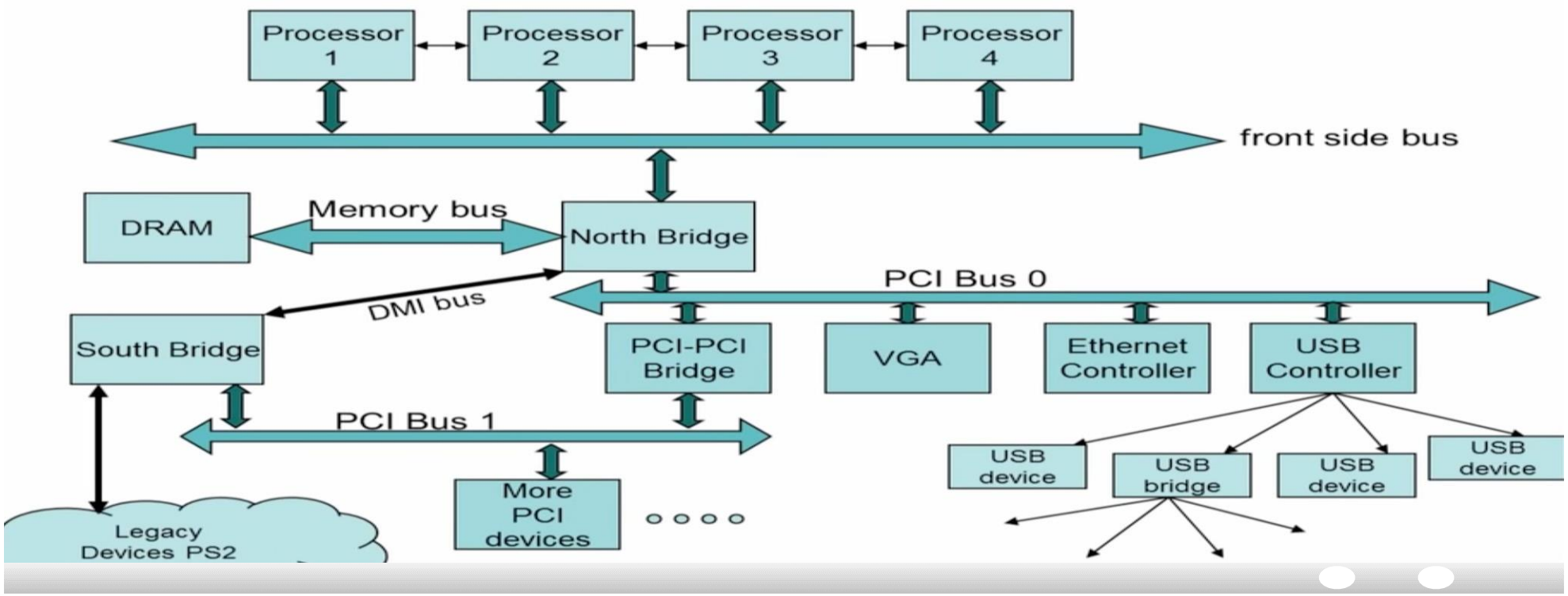
  More spaces
- Devices and RAM share the same space



Who Decides the address ranges?

- Standards / Legacy
  - Fixed for all PCs
- Plug and Play Devices
  - Address range set by BIOS or OS
  - A device address range may vary every time system is restarted

PC Organization



Thank You.